

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Packaged and deployed **Hugging Face computer vision models as containerized FastAPI APIs** on **AWS-managed Kubernetes** clusters, ensuring **GPU compatibility, CUDA driver optimization, and memory provisioning** for large-scale model inference; reduced model serving latency by **40%** through dependency management and container image optimization.
- Engineered **ML model training pipelines** collaborating with data science teams to integrate experimental computer vision workflows into production underwriting analysis systems; implemented **automated unit and integration testing** across pipeline stages, achieving **99.2% pipeline reliability** and reducing model deployment failures by **65%**.
- Architected **GPU instance workflows and EC2/GPU infrastructure provisioning** using **Docker containerization and NVIDIA driver configuration** to support concurrent model training jobs; designed **CI/CD pipelines with Jenkins** for automated model packaging, versioning, and deployment to Kubernetes, reducing deployment cycle time from **6 hours to 45 minutes**.
- Integrated model labelling workflows with existing data annotation platforms via **REST APIs and database-level data flows**; established **cloud-native architecture patterns** on **AWS (EC2, S3, Lambda)** to orchestrate labelled dataset ingestion, transformation, and model retraining; improved labelling throughput by **55%** through pipeline parallelization.
- Led **code reviews and pull request governance** on computer vision model integration code, enforcing **security threat modelling, clean code standards**, and architectural consistency across the analytics engineering team; mentored junior engineers on **modern software design patterns**, Kubernetes deployment best practices, and GPU memory optimization techniques.
- Assessed business value of emerging MLOps technologies and containerized inference frameworks using data-driven evaluation; implemented **automated testing, logging, and monitoring dashboards** for model performance tracking and production incident response; contributed technical insights through internal tech talks on GPU optimization and Kubernetes-based ML deployment architecture.

SKILLS

ML Model Deployment & Containerization – Hugging Face models, FastAPI, Docker, NVIDIA CUDA drivers, GPU provisioning, container dependency management, image optimization.

Cloud Infrastructure & Orchestration – AWS (EC2, S3, Lambda, CloudWatch), Kubernetes, managed Kubernetes services, GPU instance workflows, cloud-native architecture, infrastructure provisioning.

ML Pipeline Development & Training – Computer vision pipelines, model training orchestration, labelling workflow integration, data annotation platforms, pipeline reliability and monitoring.

CI/CD & DevOps – Jenkins pipelines, Git branching, automated testing (unit and integration), code review governance, deployment automation, model versioning.

Backend & API Development – Python, FastAPI, REST APIs, data integrations, schema design, PostgreSQL, MySQL, Oracle.

Engineering Practices – Clean code standards, security threat modelling, architectural design, mentoring, modern software design patterns, logging and monitoring.

Data & Analytics – Python (pandas, NumPy, scikit-learn), data modelling, ETL pipelines, statistical analysis, machine learning fundamentals, Power BI dashboards.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Computer Vision Underwriting Pipeline with GPU-Optimized Model Serving (HCL Technologies)

- Designed and deployed an end-to-end **Hugging Face computer vision pipeline for document classification** in underwriting workflows; packaged pre-trained transformer models as **containerized FastAPI microservices** with **CUDA-optimized inference kernels**, achieving **sub-500ms latency** on **NVIDIA GPU instances**.
- Engineered **Kubernetes deployment manifests** with automated **GPU resource requests and memory limits** to support concurrent model inference; implemented **Docker multi-stage builds** and dependency pinning to reduce container image size by **62%** and improve cold-start times across **AWS managed Kubernetes clusters**.
- Integrated model training and retraining loops with existing document labelling workflows via **REST API connectors** and **PostgreSQL data pipelines**; orchestrated dataset versioning, model artifact storage in **S3**, and automated retraining triggers using **Lambda functions**, reducing manual retraining overhead by **70%**.
- Established **Jenkins CI/CD pipelines** for model containerization, GPU compatibility validation, and blue-green deployment to production Kubernetes clusters; implemented **automated integration tests** validating GPU memory allocation, model inference accuracy, and API response latency, achieving **zero production incidents** over **18 months**.
- Architected **CloudWatch monitoring dashboards** tracking model throughput, GPU utilization, inference latency, and data drift metrics; designed **threat models** for model API security and implemented role-based access controls, maintaining **99.7% uptime SLA** and supporting **500+ daily inference requests**.

SWIFT Payments Operations Dashboard with Real-Time Message Processing (HCL Technologies)

- Developed production monitoring dashboards for a **SWIFT messaging payments-operations platform** using **Splunk and CloudWatch**, providing real-time visibility into **MT/MX message flows, ISO 20022 transaction processing, and SEPA cross-border payment routing**; reduced payment message incident response time by **50%** through automated alerting on SLA breaches.
- Engineered **Python ETL pipelines** for reconciliation workflows and payment transaction data validation across **Oracle and PostgreSQL** systems; implemented **scheduled reporting** and **Power BI dashboards** tracking daily transaction volumes, settlement status, and regulatory compliance metrics for banking operations teams.
- Collaborated with banking-domain business analysts and compliance teams to implement **GDPR data protection controls**, audit trail logging, and **PSD2 Open Banking** compliance checks within payment processing workflows; coordinated UAT cycles and post-implementation triage across interconnected ERP and messaging systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)

- Certificate of Completion in Python programming from **University of Michigan** (Coursera)